

Weather Intelligence & Environmental Risk Upgrade Walkthrough

Overview

CropGuardian has evolved from a pure disease detection system into a **Weather-Aware AI Crop Disease Risk & Advisory Platform**. The Google ADK architecture has been expanded to intelligently synthesize visual disease data with real-time geospatial weather conditions.

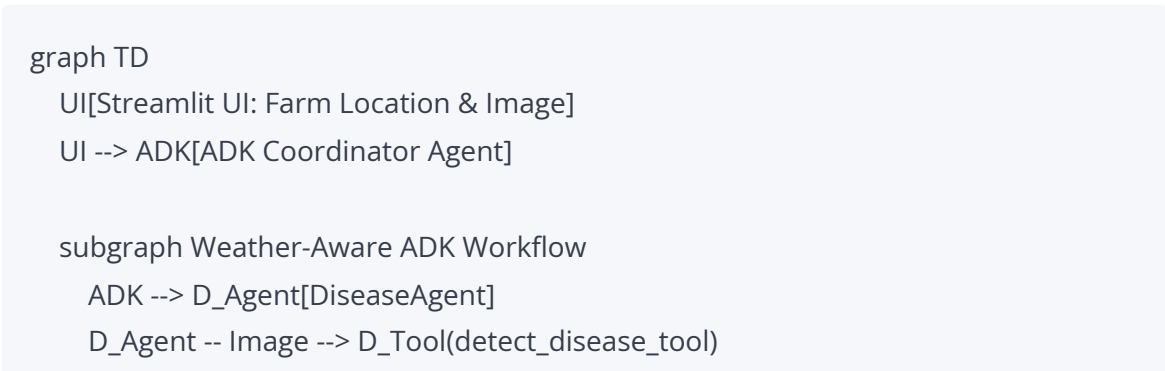
What's New?

1. **WeatherAgent**: Integrates with Open-Meteo to transparently geocode user location inputs (City, State) or exact coordinates (Lat/Lon). It retrieves high-precision meteorological data (temp, humidity, rain, wind).
2. **EnvironmentalRiskAgent**: Evaluates the retrieved weather against the diagnosed disease. Utilizing deterministic, botanical heuristics, it calculates fungal, bacterial, and heat stress risks to compute an overall spread probability.
3. **Upgraded UI & Advisory**: The Streamlit interface now provides actionable Environmental Intelligence dashboards, and the Gemini-powered Advisory Agent issues precise Weather-Based Warnings and Immediate Actions .

1. Updated ADK Workflow Diagram

The workflow forces strict chronological state passing:

Disease -> Weather -> Severity -> Env Risk -> Advisory



```

D_Tool -.-> Legacy_D[DiseaseDetectionAgent]

D_Tool --> W_Agent[WeatherAgent]
W_Agent -- Location --> W_Tool(get_weather_tool)
W_Tool -.-> Legacy_W[WeatherAgent]

W_Tool --> S_Agent[SeverityAgent]
S_Agent -- Disease + Image --> S_Tool(analyze_severity_tool)
S_Tool -.-> Legacy_S[SeverityAgent]

S_Tool --> R_Agent[EnvironmentalRiskAgent]
R_Agent -- Disease + Weather --> R_Tool(assess_risk_tool)
R_Tool -.-> Legacy_R[EnvironmentalRiskAgent]

R_Tool --> A_Agent[AdvisoryAgent]
A_Agent -- Disease + Severity + Weather + Risk --> A_Tool(generate_advice_tool)
A_Tool -.-> Legacy_A[AdvisoryAgent]
end

Legacy_A -.-> ADK
ADK --> | Unified Weather & Disease JSON | UI

```

2. Updated Project Structure

The agents folder has expanded cleanly:

```

agricare/
  agents/
    adk_workflow/      # ADK Orchestration
    advisory_agent/    # Gemini LLM Integration
    coordinator_agent/ # Legacy Fallback Orchestrator
    disease_detection_agent/ # Visual ML Model
    environmental_risk_agent/ # [NEW] Deterministic Risk Scoring
    feedback_agent/    # Model Retraining Feedback
    severity_agent/    # Visual Severity Scoring
    weather_agent/     # [NEW] Open-Meteo Integration
  app/
    pages/
      1_Disease_Detection.py # [MODIFIED] Weather Dashboards

```

```
main.py
tests/
test_weather_agent.py      # [NEW]
test_environmental_risk_agent.py  # [NEW]
```

3. Weather Intelligence Walkthrough

When a user visits the Streamlit app:

1. They encounter a new **Farm Location** input block. They can choose to input City + State + Country (e.g., "Cherthala, Kerala, India") or raw Latitude + Longitude .
 2. Behind the scenes, the WeatherAgent intercepts this, queries the geocoding-api.open-meteo.com to resolve coordinates silently, and then hits api.open-meteo.com .
 3. The exact resolved location is appended to the payload ("latitude": 9.6848, "longitude": 76.3356) to aid in debugging, mapping, and report generation.
 4. The UI renders this via 4 crisp metric cards (Temp, Humidity, Precipitation, Wind) directly above the risk assessment.
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4. Environmental Risk Scoring Explanation

[!IMPORTANT]

The EnvironmentalRiskAgent is strictly deterministic. It does NOT use LLM reasoning for threshold calculations, making the system highly predictable and scientifically sound.

How it works:

- **Fungal Risk:** Evaluated strictly on humidity. If humidity > 85%, Fungal Risk is marked as Very High (score: 90) because spores rapidly germinate.
- **Bacterial Risk:** Evaluated on a matrix of humidity (>80%) and warm temperatures (25-35°C), or simply rainfall (which causes bacterial splash dispersal).
- **Heat Stress Risk:** Evaluated based on thermal maximums (temp >= 35°C).
- **Spread Probability:** The highest base score (Fungal, Bacterial, or Heat) is taken as the baseline. If an active disease is already detected, a +10% penalty is applied to the spread likelihood. If the plant is healthy, a -20% reduction is applied.
- **Concept separation:** This provides a distinct metric separating Severity ("How badly damaged is the leaf right now?") from Environmental Risk ("How likely is the

farm to lose the crop tomorrow?").